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Multiclass skin lesion classification and localization from dermoscopic images using a novel network-level fused deep architecture and explainable artificial intelligence

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Abstract

Background and objective Early detection and classification of skin cancer are critical for improving patient outcomes. Dermoscopic image analysis using Computer-Aided Diagnostics (CAD) is a powerful tool to assist dermatologists in identifying and classifying skin lesions. Traditional machine learning models require extensive feature engineering, which is time-consuming and less effective in handling complex data like skin lesions. This study proposes a deep learning-based network-level fusion architecture that integrates multiple deep models to enhance the classification and localization of skin lesions in dermoscopic images. The goal is to address challenges like irregular lesion shapes, inter-class similarities, and class imbalances while providing explainability through artificial intelligence.

Methods A novel hybrid contrast enhancement technique was applied for pre-processing and dataset augmentation. Two deep learning models, a 5-block inverted residual network and a 6-block inverted bottleneck network, were designed and fused at the network level using a depth concatenation approach. The models were trained using Bayesian optimization for hyperparameter tuning. Feature extraction was performed with a global average pooling layer, and shallow neural networks were used for final classification. Explainable AI techniques, including LIME, were used to interpret model predictions and localize lesion regions. Experiments were conducted on two publicly available datasets, HAM10000 and ISIC2018, which were split into training and testing sets.

Results The proposed fused architecture achieved high classification accuracy, with results of 91.3% and 90.7% on the HAM10000 and ISIC2018 datasets, respectively. Sensitivity, precision, and F1-scores were significantly improved after data augmentation, with precision rates of up to 90.91%. The explainable AI component effectively localized lesion areas with high confidence, enhancing the model's interpretability.

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Conclusions The network-level fusion architecture combined with explainable AI techniques significantly improved the classification and localization of skin lesions. The augmentation and contrast enhancement processes enhanced lesion visibility, while fusion of models optimized classification accuracy. This approach shows potential for implementation in CAD systems for skin cancer diagnosis, although future work is required to address the limitations of computational resource requirements and training time.

Clinical trial number Not applicable.

Keywords Dermoscopy, Skin cancer, Deep learning, Network-level fusion, Shallow neural network, Explainable artificial intelligence

Introduction

The largest organ in the body, the human skin, is impacted by many things, such as exposure to ultraviolet (UV) radiation, sunburn, lifestyle decisions, alcohol and smoke usage, physical activity, infections, and work situations [1, 2]. The human skin comprises three main layers: the epidermis, dermis, and hypodermis [3]. Melanocytes, which make up the epidermis, can synthesize melanin. Under certain conditions, such as exposure to ultraviolet (UV) light, the melanin production in the epidermis is abnormally elevated [4].

Approximately 123 million new skin cancer cases are identified annually [5]. Early diagnosis is effective in treating the majority of skin malignancies [6]. Aside from the illness and distress caused by skin disorders, skin cancer can be fatal if treatment is ignored. Skin cancer is caused by the uncontrolled proliferation of aberrant cells in the skin [7]. According to estimations provided by the American Cancer Society [8], it is projected that there will be an estimated 1.9 million extra cases of cancer detected in the year 2021. In 2021, an estimated 1670 Americans would succumb to skin cancer daily. Research indicates that skin cancer is the most sophisticated kind of cancer in terms of diagnosis and prognosis, and it is also one of the most common types of cancer in developing countries [9]. The United States has reported 500,000 new cases, making it the 19th most prevalent cancer globally [6].

There are two distinct forms of skin cancer: non-melanoma and melanoma [10]. Melanoma, also known as aberrant melanocyte proliferation, is the origin of malignant cancers [11]. Several kinds of skin cancer, such as Squamous cell carcinoma [12], basal cell carcinoma [13], and Merkel cell carcinoma [14], are distinct from melanoma. Melanocytes, the cellular entities responsible for pigment production, serve as the origin of melanoma. If not promptly recognized and appropriately treated, it can be pretty lethal [7]. It has a 93% five-year survival rate when discovered early. When detected early, it exhibits a five-year survival rate of 93%. However, in the event of metastasis to far lymph nodes [15] and other organs, the proportion may decrease to 27%. Skin cancer can be diagnosed with dermoscopy, a non-invasive method

that provides high-resolution skin scans. Dermatologists have a fantastic opportunity to use dermoscopy to obtain high-resolution magnified images. This approach dramatically improves the visual quality of the collected data [16]. However, the existence or lack of additional symptoms or signs, as well as the various abnormalities' size, shape, colour, edge, boundary, and position, make dermoscopy diagnosis of skin cancer challenging [17, 18].

The manual evaluation of dermoscopic images hinders a more precise detection of skin cancer [19]. The degree of accuracy with which the lesion is diagnosed depends on the dermatologist's experience [20]. There are few alternatives to macroscopic and biopsy [21] procedures for diagnosing skin cancer. In medical imaging, several researchers have developed computer-based detection (CAD) [22] techniques for skin cancer [23]. Finding the lesion and segmenting the image is the initial step in categorizing skin lesion images [16, 24]. Numerous studies have been conducted in this area because there is a wide diversity of lesions, making it challenging to identify and segment them [25]. The CAD system utilizes image processing and machine learning algorithms [26]. Dermatologists have used the CAD [19] approach to identify skin lesions for over three decades. Computer-aided diagnostic (CAD) can potentially enhance decision-making for dermatologists, especially those with limited expertise [27].

The fundamental components of a computer-aided design (CAD) system for the detection and classification of skin cancer include (a) the preprocessing of images [28] (b) skin lesion segmentation [29], (c) feature extraction [30] and selection [31], and (d) training a classifier for decision-making using advanced machine learning techniques. Each phase possesses inherent significance and is subject to the effect of the preceding steps. In addition, improved segmentation and feature extraction steps improve the accuracy of classification [30].

Identifying skin cancer in medical imaging presents several challenges, including considerations of colour, texture, form, hair, and resemblance among lesions [32]. The clinical appearances of pigment skin diseases exhibit significant variability during their initial and advanced stages, accompanied by a notable level of variance within

each class and commonalities across different classes. The primary challenges in identifying skin lesions are as follows: i) Early-stage lesions exhibit a relatively uniform hue, borders, and abnormal appearance [33], but later-stage lesions display a wide range of diverse appearances. As a result, it is difficult to determine whether initial skin lesions are benign or malignant with the naked eye [33].

Despite this, several computerized techniques have been introduced in the literature for identifying and classifying skin lesions. The techniques that have already been presented are based on traditional methods such as clustering algorithms, handcrafted features, and semi-supervised learning. The conventional methods focused on the lesion shape and location, but that technique did not work for the diverse image nature dataset. Therefore, deep learning-based methods showed considerable success in medical imaging for lesion localization and classification with a greater precision rate. The performance of deep learning models is based on the accurate and handsome amount of labelled data. Training a deep learning model using low contrast and noisy data impacts the model's accuracy.

Over the past few years, feature-level fusion has been extensively studied and implemented to improve classification accuracy, but it usually accompanies higher model dimensionality, increased training complexity, and duplicate or repetitive information which leads to higher costs and potential performance degradation [34] (other domain like agriculture and remote sensing). In this work, we proposed a network-level fusion deep learning architecture for accurately classifying and localizing skin lesions.

Deep learning-based image processing techniques are becoming increasingly popular [35], particularly in medical imaging, as deep learning technology advances. Many computerized techniques based on deep learning have been introduced in the literature for the classification and localization of skin cancer [36]. The most utilized CNN architectures for the classification process are VGG16, ResNet50, DenseNet, and InceptionV3. Also, many researchers introduced single-stage customized architectures for lesion classification. Karar et al. [37] demonstrate how difficult it is to classify skin lesions and provide a strategy for diagnosing them. They created an initial image pipeline that involved removing hairs from images, enhancing the dataset, and resizing images based on each model's requirements. They learned the EfficientNets B0-B7 on the HAM10000 dataset using transfer learning on pre-trained ImageNet weights and fine-tuning the Convolutional Neural Networks. Based on the results, the EfficientNet B4 architecture obtained better results, such as an F1-Score value of 87% and 87.91% accuracy.

Muhammad et al. [38] described a framework for classifying and segmenting skin lesions. The method being given consists of two steps. First, an architecture based on a (MASK R-CNN) was used for lesion segmentation. To execute activation based on the visualization of higher features, a 24-layered CNN model was developed for the classification phase. The Resnet-50 model was utilized for both segmentation and classification. Three datasets, including PH2, ISBI2016, and ISIC2017, were used to validate the segmentation procedure, while the HAM10000 dataset was used for the classification. Based on the chosen set of metrics, the presented method obtained a sensitivity (85.57%), precision (87.01%), F1-Score (86.28%), and accuracy (86.5%), respectively. Arshad et al. [39] presented a deep learning-based computer-aided diagnosis system for several classes of skin lesion classification. The ResNet-50 and ResNet-101 models are chosen as pre-trained models. Both refined deep models were trained on augmented datasets using transfer learning. After that, features were extracted, and fusion was carried out via an altered serial-based methodology. Finally, the optimal features were chosen using the skewness-controlled SVR technique to improve the accuracy. They used HAM10000 dataset for the experimental process and shows the improved accuracy of 91.7%. Muhammad et al. [40] focused on lesion segmentation, which was challenging due to the region's wide range of colors and textures. Another issue is that cutaneous melanoma was detected and treated early enough; it can still be cured. The author presented two main steps: segmentation and classification. They made two CNN architectures and generated the binary images that are further fused using a joint probability distribution and marginal distribution functions. An architecture of a 30-layered CNN was constructed and trained on the HAM10000 dataset for lesion classification. The presented framework was evaluated on five skin datasets (ISBI2016, ISIC2017, ISBI2018, ISIC2019, and HAM10000) and obtained accuracy of 92.06%, 92.68%, and 92.70%, 86.4%, and 87.02%, respectively. Bharat et al. [41] introduced an automated computer-aided skin lesion detection and classification system. They presented SLDCNet, a hybrid preprocessing-based transfer learning model for SLDC. The hybrid Gaussian filter (HGF) with a connected component label (CCL) based rapid march inpainting approach was first employed for hair removal and skin lesion denoising. The cancer location was then identified using a segmentation technique based on full-resolution convolutional networks (FrCNs). Deep residual learning extracted the features, and improved classification accuracy was obtained.

Akhilesh et al. [2] emphasize that early detection and classification of malignant growths on the skin can be costly and complex. The CNN and recurrent networks are examples of deep learning architectures constructed

for non-handcrafted complicated feature extraction. This study introduced a cascaded ensemble model that combines a multi-layer perceptron based on handcrafted features with ConvNet integration to further enhance the effectiveness of the ConvNet models. The experimental process shows that the presented ensemble deep learning model's accuracy increased by 98.3% from 85.3%. Ranpreet et al. [42] presented a new CNN architecture based on several convolution building blocks. They fine-tuned the network in layers and hyperparameters for improved accuracy. Sumit et al. [43] presented a modified deep-learning model with fuzzy logic-based image segmentation to diagnose skin cancer. They updated You Look Only Once (YOLO) deep neural network model for better detection results.

In summary, the works mentioned above focused on lesion classification and detection based on some custom and pre-trained models. Few combine the CNN and handcrafted features for improved accuracy and precision rate. However, they did not focus on the self-attention module to extract in-depth information about an image. Also, they did not work on the network-level fusion. In this work, we presented a new network-level fusion architecture with explainable AI for skin lesion classification and localization.

The significant contributions of our work are as follows:

- Implemented a hybrid technique based on the Deep Denoisy Network and haze reduction for contrast enhancement and dataset augmentation.

- Proposed a 5-block inverted residual deep learning architecture with multiple scaling factors and fixed stride.
- Design a 6-block inverted bottleneck deep learning architecture with convolution filter sizes 1×1 and 3×3 for in-depth information extraction.
- Fused both designed models using a network-level fusion approach named depth concatenation instead of features level fusion.
- Performed a detailed ablation study to analyze the performance of each middle step and compared it with recent state-of-the-art (SOTA) techniques.

Methods

In this section, the proposed methodology for skin lesion classification and localization has been presented. The proposed work is based on network-level deep architecture fusion and explainable AI. Figure 1 shows the proposed architecture, which includes three essential steps. In the first step, a contrast enhancement technique was used to improve the lesion region, and then data augmentation was performed. In the second step, two CNN models were proposed based on a 5-block inverted residual and a 6-block inverted bottleneck. In particular, the 5-block inverted residual model is implemented with the usage of grouped convolution, fewer depth-wise filters, and also in paired skip connections that allow for the retention of fine-grained features typical for dermoscopic analysis. Similarly, the 6-block inverted bottleneck model has a separate channel expansion-compression scheme

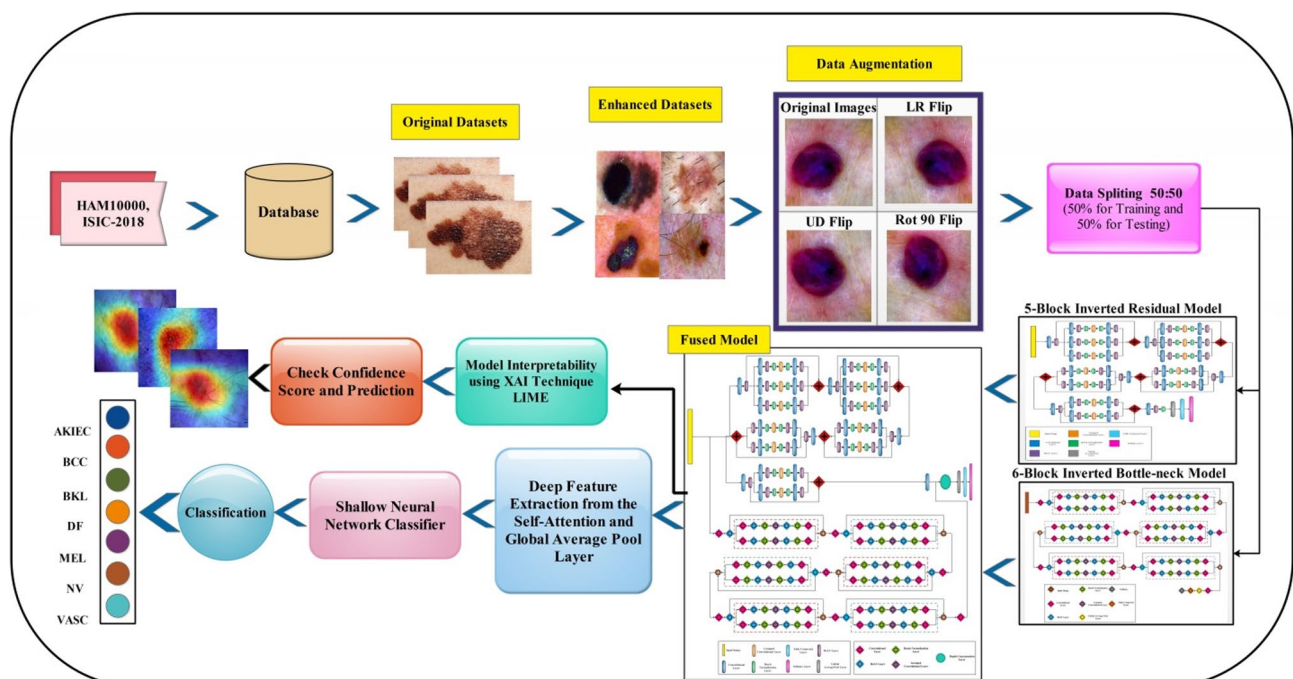


Fig. 1 Proposed architecture of multiclass skin lesion classification and localization

that differs but is based on the channel expansion-compression scheme developed for MobileNetV2 and modified with grouped convolution layers and deeper residual paths. The designed models are then fused using a depth-concatenation layer and performed training. The Bayesian Optimization technique was employed in the training process to initialize the hyperparameters. Deep features are extracted from the trained fused model, and classification is performed. In the classification, shallow neural networks that contain one or two hidden layers have been employed for the final classification results. In parallel, the proposed fusion network has been interpreted using an explainable AI technique named LIME for lesion region localization and prediction. A detailed description of each step is given below.

Datasets of this work

HAM10000: The proposed methodology is validated on HAM-10,000 [40] dermoscopy images. The dataset comprises 10,015 images of skin diseases, with the dataset's class description presented in the data augmentation section. The RGB images are saved as JPEG files. Two distinct subsets comprise the entire dataset: training and testing. The 50% of the data is assigned for training, while the other 50% is used for testing. Basal Cell Carcinoma (BCC), Melanocytic Nevi (NV), Dermatofibroma (DF), Melanoma (MEL), Benign Keratosis-like Lesions (BKL), Vascular Lesions (VASC), and Basal Cell Carcinoma (BCC) are the seven classes comprising the dataset [44]. Figure 2 shows the sample images of the HAM10000 dataset.

ISIC2018: International skin imaging collaboration ISIC is a melanoma detection organization. ISIC2018 [45] dataset currently consists of more than 10,015 dermoscopy images organized into seven classes, one of which analyzes skin lesions for melanoma detection [46]. The description of the number of images is given in the augmentation section. A few sample images of this dataset are shown in Fig. 2.

Contribution 1: Lesion contrast enhancement

Contrast enhancement is important in computer vision applications to improve image pixels for better visibility. In medical imaging, contrast enhancement aims to improve the infected or cancer region for better feature extraction and localization. In this work, we proposed a hybrid contrast enhancement technique based on a deep denoising network and haze reduction to remove the speckles and enhancing the fine-grained texture features without blurring the vital lesion boundaries.

Denoising with DnCNN

Convolutional neural network (CNN) for denoising DnCNN is capable of doing Gaussian denoising on noise

that is not known. The DnCNN model is well-suited for addressing speckle noises characterized by unpredictable pixel noise levels. The network architecture of the DnCNN model is constructed by adapting the VGG network for image denoising and adjusting its depth according to the most advanced denoising patch sizes available [47]. To improve despeckling performance, the SRAD filter output image is sent directly to DnCNN due to CNN's deep learning capabilities. DnCNN learns features from the noisy image and estimates the original image using the retrieved features. This network employs Batch Normalization (BN) and Rectifier Linear Unit. The training data includes 10,015 HAM10000 natural images and 10,015 ISIC2018 images. The dataset images underwent denoising using a 20-layer Convolutional Neural Network (CNN). The process of estimating speckle noise involves doing a residual mapping, denoted as $\mathfrak{R}(a, b)$, which provides an accurate estimation of speckles, denoted as $\mathfrak{R}(a, b) \approx S_e(a, b)$. Subsequently, the estimation of the original image is obtained by Eq. 1.

$$\mathcal{F}_i(m, n) = \frac{\mathfrak{l}(a, b)}{\mathfrak{R}(a, b)} \quad (1)$$

where, $\mathcal{F}_i(a, b)$ is an estimate of $\mathcal{F}(a, b)$. In terms of the loss function ($\mathcal{L}_f$), the average mean squared error (MSE) corresponding to the residual image and the estimate of noise may be defined [48] by Eq. 2.

$$\mathcal{L}_f(\omega) = \frac{1}{2}X \sum_{i=1}^X \|\mathfrak{R}(\mathfrak{l}_i; \omega) - (\mathfrak{l}_i - \mathcal{F}_i)\|_F^2 \quad (2)$$

The set X consists of pairs of original and noisy images, while ω represents the training parameter of the deep network, which can be acquired by utilizing the loss function. The network structure is discussed below.

Conv + ReLU In the initial layer, 64 filters with dimensions of $3 \times 3 \times C$ are employed to produce 64 feature maps. Subsequently, rectified linear units $ReLU, \max(0, \cdot)$ are employed to introduce nonlinearity. The variable C denotes the number of image channels, namely $c = 1$ for a grayscale image and $c = 3$ for a colorized image.

Conv + BN + ReLU For layers $2 \sim (D - 1)$, a total of 64 filters with dimensions of $3 \times 3 \times 64$ are employed. Additionally, batch normalization is incorporated between the convolution and ReLU operations.

Conv In the last layer, the output is reconstructed using C filters with a size of $3 \times 3 \times 64$.

This network is trained in two stages: the network design, which is based on a modified model, and the learning model, which is derived from the residual learning

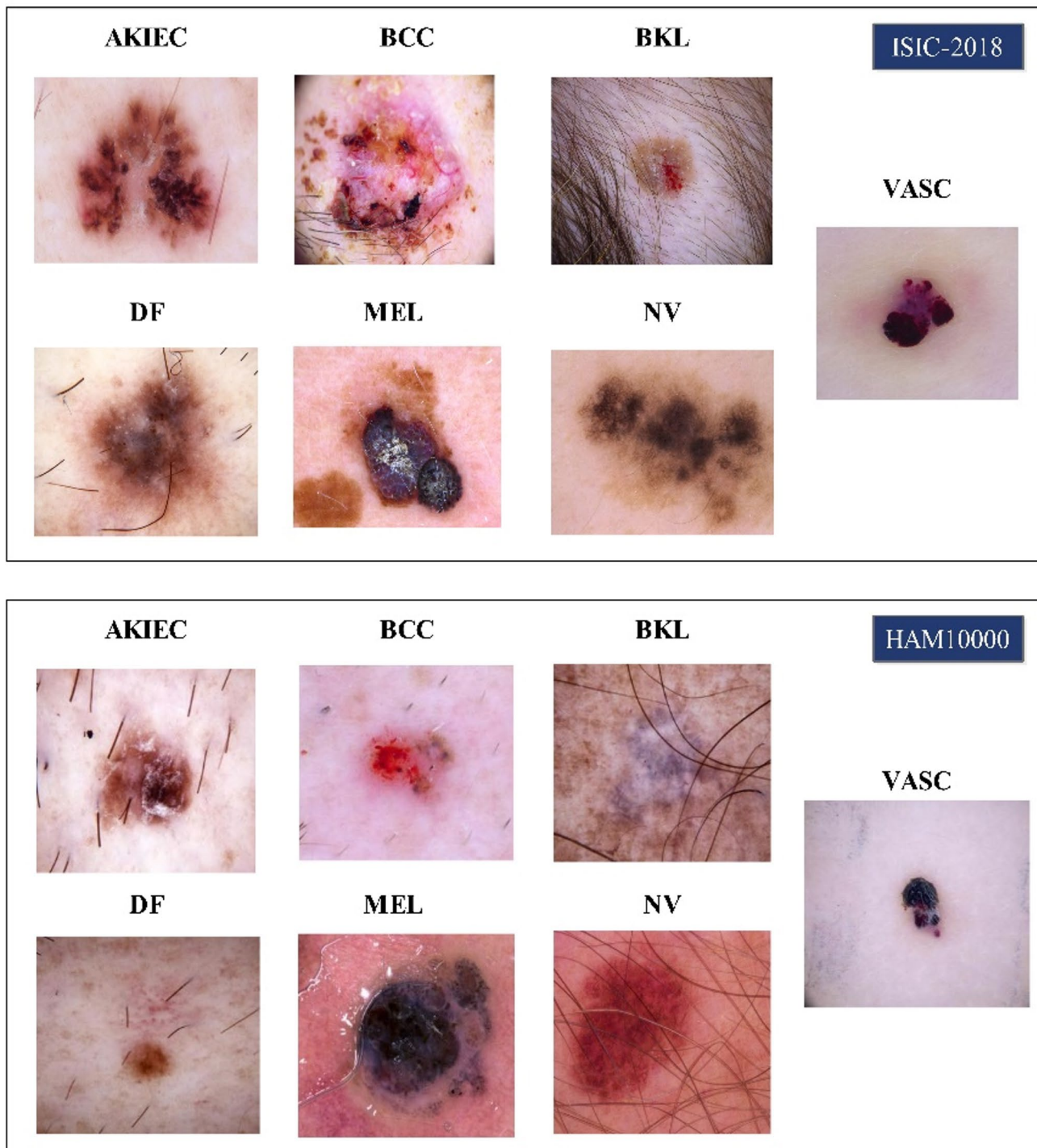


Fig. 2 Sample images of ISIC2018 and HAM10000 dataset

formulation. All pooling layers are eliminated to accelerate computation, and convolutional filter sizes are set to 3×3 . Consequently, the DnCNN's receptive field at a depth of D_{pth} will be $(2d + 1) \times (2d + 1)$. A bigger receptive field size results from increasing and thus expands the picture region's context information. Therefore, it is crucial to choose an optimal value for d to

achieve a balanced compromise between the efficiency and effectiveness of DnCNN. To achieve this objective, the experimental determination of the depth D_{pth} for DnCNN is set at 20.

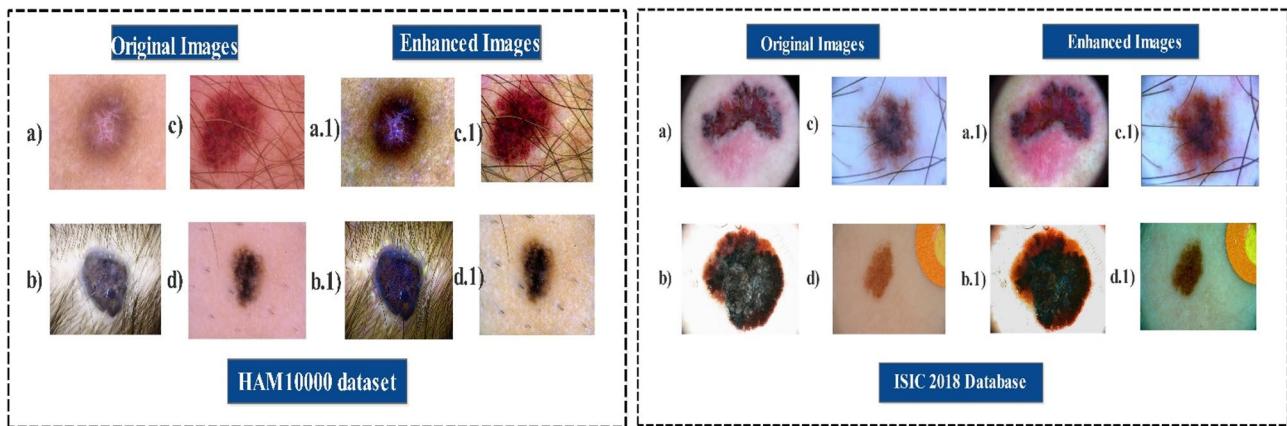


Fig. 3 Original images to enhance images by applying haze reduction on both dataset HAM-10,000 and ISIC-2018 (a),(b),(c), and (d) show the original image and (a.1), (b.1), (c.1) and (d.1) show the enhance images

Sr.No.	Disease Name	No of Images	Augmented Images	For Training and Testing 50:50
1.	Actinic keratoses (AKIEC)	327	5242	2621
2.	Basal cell carcinoma (BCC)	514	5225	2613
3.	Benign keratosis-like lesions (BKL)	1099	5852	2426
4.	Dermatofibroma (DF)	115	4680	2340
5.	Melanoma (MEL)	1113	5423	2712
6.	Melanocytic Nevi (NV)	6705	6705	3353
7.	Vascular Lesions (VASC)	142	4544	2272

Fig. 4 HAM10000 dataset before and after augmentation

Haze reduction with denoisy network

After the DcNN Denoisy network, haze reduction has been performed based on the following mathematical model [49]:

$$I_m(x) = R_s(x) T_m(x) + L_a(1 - T_m(x)) \quad (3)$$

The variable I_o represent the observed intensity, R_s denotes the scene radiance, L_a denotes the atmospheric light and T_m denotes the transmission map that indicates the amount of light reaching the camera. The dehazed image R_s is obtained via dehazing algorithms using an assessment of the atmospheric light and transmission map, as follows [49]:

$$R_s(x) = (I_m(x) - T_m(x) L_a) / (\max(T_m(x), T_0)) + L_a \quad (4)$$

Five phases are involved in the dehazing algorithm as follows [49]: 1) Calculate the atmospheric light L_a using a previous on a dark channel; ii) T_m , the estimated transmission map; iii) Improve the transmission map estimation; iv) Restore the image; v) Apply contrast enhancement if desired. A few results of the contrast enhancement technique are shown in Fig. 3 for both selected datasets. This figure shows that the infected region pixel contrast is improved after employing the proposed hybrid technique.

Dataset augmentation As shown in Fig. 4, the number of images in each class is highly imbalanced; therefore, an

augmentation is required before training a CNN model. In this work, we performed three operations for the generations of newly labelled data, such as rotate 90, right-left flip (LR), and up-and-down flip (UD). The mathematical formulation of these operations are.

Let the input image is presented by the $g_{(\alpha, \beta)}$ with the size of $w \times r$, where w denoted the row pixels $\{\alpha \in (1, 2, 3, \dots, x)\}$ and r presented the column pixels $\{\beta \in (1, 2, 3, \dots, y)\}$. The performed operations are flip left, flip right and rotation 90.

$$g_{(\alpha, \beta)}^{\rightarrow} = g_{\alpha}(x + 1 - r) \quad (5)$$

$$g_{(\alpha, \beta)}^{\leftarrow} = g_{\beta}(y + 1 - w) \quad (6)$$

$$g_{(\alpha, \beta)}^{90} = \begin{bmatrix} \cos 90 & -\sin 90 \\ \sin 90 & \cos 90 \end{bmatrix} \begin{bmatrix} g_{\alpha} \\ g_{\beta} \end{bmatrix} \quad (7)$$

Where $g_{(\alpha, \beta)}^{\rightarrow}$ presented the flip left, $g_{(\alpha, \beta)}^{\leftarrow}$ presented the flip right, and $g_{(\alpha, \beta)}^{90}$ denoted the rotation 90. We repeated this operation until 5000 images in each class were generated. Hence, the newly generated dataset includes 35,000 enhanced images compared to 10,015 in the older dataset (HAM10000 and ISIC2018).

Proposed 5-block inverted residual model (IBR⁵Net)

The proposed IBR5Net Model has been presented in this section. The design model consists of 101 layers, which are inverted residuals. The IBR5Net architecture is inspired to have a low representational capacity while requiring less parameters and serving as a lower-cost model. Because of this, it can capture low to mid-level features in an efficient manner. The overall architecture of the proposed model is shown in Fig. 5. The dimension of the initial input layer is 227 by 227, with a depth of 3. After that, a convolutional layer is the first layer, with a stride of two, a filter size of 3×3 , and a depth of 16. Subsequently, the ReLU layer, also referred to as the activation layer, was added. The main residual block comprises a convolution layer, batch normalization layer (BN), ReLU layer, and grouped convolutional layer in the parallel structure.

The first residual block employs a 1×1 filter size convolutional layer of depth 32, stride 1, and batch normalized ReLU layer. The BN layer employs a grouped convolutional layer with a filter size 3×3 and a channel size of 1. Apply a 1×1 filter, depth 16, stride 1, and activation ReLU layer after the third convolutional layer. All leftover blocks have one stride. The parallel block is prepared for the following parallel blocks layer by layer. A parallel block with inverted residuals consists of three separate channels with a skip link. The main purpose of parallel block addition is to get more in-depth information about an image.

The next layer is a convolutional layer with a 3×3 filter size, 32 depth size, and stride value of 2. Following that, the ReLU activation layer has been added. The second residual block shows a similar method. The leftover block is layer-based. The two convolutional layers are 64 and 32 deep. The convolutional layer has a 1×1 filter with stride 1. The above procedure is iterated for the paralleled block utilizing extra layers to include parallel residual blocks. A parallel block with inverted residuals has three skip-linked channels.

The convolutional layer uses a 3×3 filter with 64 depth and a stride value 2. A ReLU activation layer is added after each convolutional layer. Similar processes occur in the third residual block. The third residual block has two convolutional layers of depth sizes 128 and 64. The convolutional layer has a 1×1 filter with stride (1) The above procedure is iterated for the paralleled block utilizing extra layers to include parallel residual blocks. Two channels with skip links form a parallel block with inverted residuals. The next layer is a 3×3 convolutional layer with 128 depth and a stride value (2) The fourth residual block has the same pattern as the previous blocks. A 128-depth convolutional layer follows the 256-depth. This convolutional layer uses a 1×1 filter with stride 1. For parallel residual blocks, the above procedure is iterated with extra layers. After a 3×3 filter and 256-depth convolutional layer, a ReLU activation layer has been added. Convolutional layers, ReLU activation, BN, and grouped convolutional layers have been included in a parallel path for the fifth residual block. The fifth residual block has a constant layer structure with 512- and 256-depth convolutional layers. The convolutional layer has one stride and a 1×1 filter size.

The last layer employs a convolutional layer with a stride value 2, 3×3 filter size, and 512-depth size. Following the ReLU operator, batch normalization is implemented. A global average pooling layer decreases the dimensionality of a neural network before classification. The feature maps are condensed into a single value for each channel. The output is flattened by a fully connected layer, followed by a softmax layer for categorization. The designed model consists of 101 layers and 8.4 million learnables.

Proposed 6-block inverted bottleneck model (IBR⁶Net)

The 6-block inverted bottleneck architecture is implemented to extract more abstract, and deeper features. The goal of the 6-block specific architecture is to expand the feature space, capture deeper features, and compress the model that contains many more rich interactive information, while still using fewer parameters. The proposed inverted bottleneck model consists of several layers in a bottleneck fashion, such as decreasing and expanding each layer's dimension. The proposed

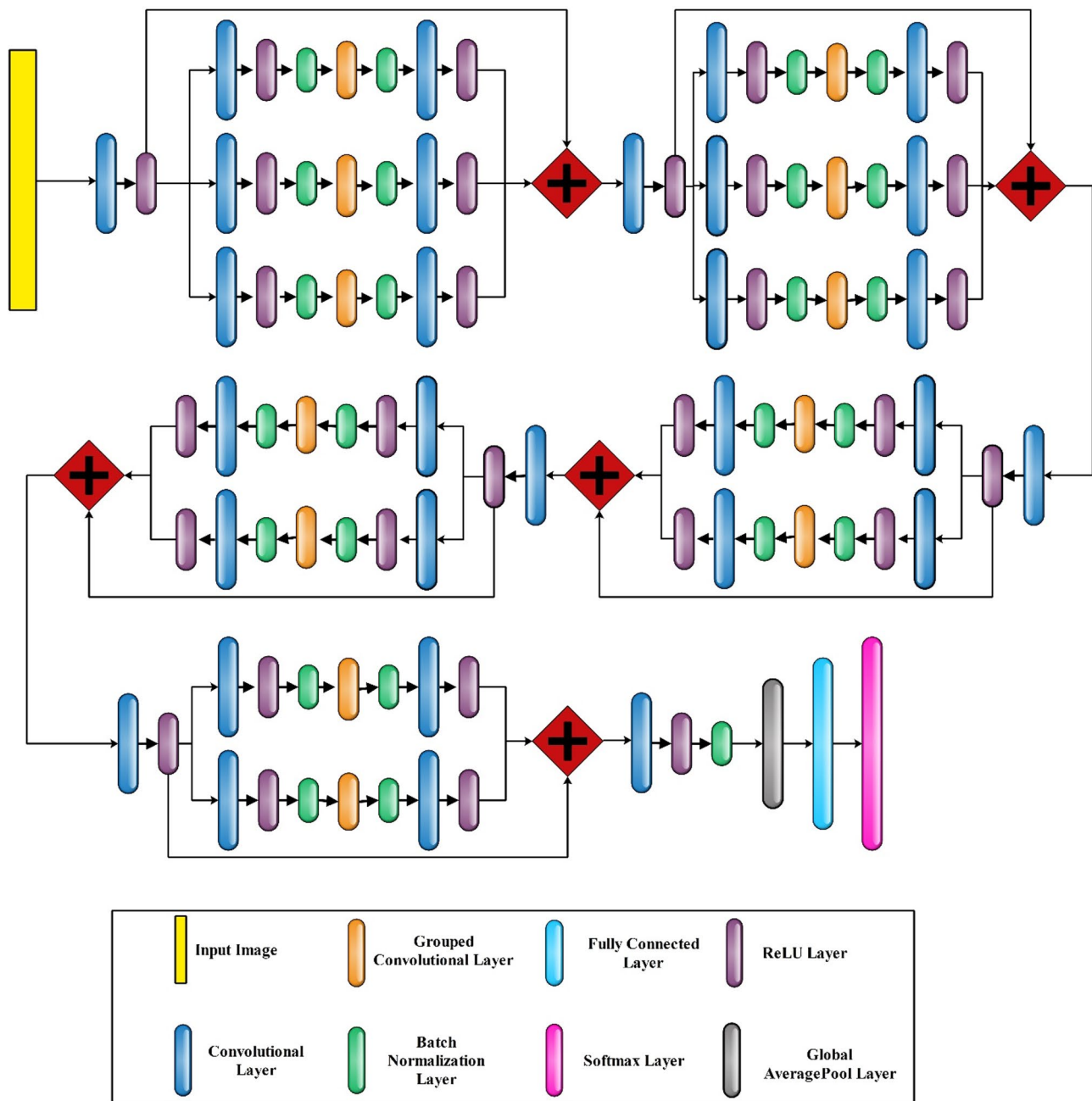


Fig. 5 Proposed layered model structure using 5-block inverted residual design

architecture comprises three operations: a depth-wise separate convolution to analyze spatial input, a 1×1 convolution to reduce the number of channels, and a further 1×1 convolution to raise the number of channels again. The inverted bottleneck reduces channels, expands them first, conducts a mild depth-wise convolution, and then shrinks them back to their original size. Prioritizing channel expansion allows the model to get richer features before applying depth-wise convolution, improving performance at a cheaper computational cost. Thus, for environments with restricted resources, the inverted

bottleneck design has become an essential part of developing deep learning models that are both successful and efficient. A particular bottleneck block consists of 3 main parts:

1×1 Convolution layer In bottleneck designs, the filter size of the 1×1 convolutional layer reduces channel size for effectiveness and offers nonlinearity for enhanced expressiveness. This layer helps balance computational cost and performance, which is crucial for designing efficient deep neural networks.

3×3 grouped convolution layer The efficient spatial feature capture in bottleneck designs is attributed to the 3×3 grouped convolutional layer, which follows the filter size of the 3×3 convolutional layer. It improves feature extraction by aggregating input channels to allow for parallel processing. By balancing computing efficiency with model expressiveness, this design choice optimizes outcomes for deep learning tasks.

1×1 Convolution layer Reduce channel size and fine-tune feature integration to accelerate computation. In bottleneck topologies, 1×1 convolutional layer filter size occurs after 3×3 grouped convolution filter size. By balancing model expressiveness and efficiency, this design choice optimizes performance in deep learning tasks.

6-block inverted bottleneck architecture

The proposed 6-block inverted residual bottleneck architecture comprises 107 layers and 9.1 million learnables. This network accepts an input of size $227 \times 227 \times 3$. The overall network architecture is illustrated in Fig. 6. The first layer is a convolutional layer having a filter size of 3×3 , a depth of 16, and a stride of 2. The ReLU layer was

added after each convolutional layer. A parallel route, comprising a convolution layer, a ReLU layer, BN, and a grouped convolutional layer, is present in the first residual block.

The main residual block has a parallel approach with a convolution, BN, ReLU, and grouped convolutional layers. First, a 1×1 filter size convolutional layer at depth 32 (stride 1) is added. A ReLU layer, BN, and grouped convolutional layer with a 3×3 filter and one channel follow. In the third stage, a 1×1 convolutional layer with 16 depth and a stride of 1 is added. The stride remains one in each leftover block. Before inserting the parallel residual block, more layers are added. A parallel block with an inverted residual bottleneck has two pathways and one skip link.

Next is the convolutional layer, with a stride value of 2, 3×3 filter size, and 32 depth. The second residual block follows a layer pattern. Two convolutional layers have depths of 64 and 32, respectively. This convolutional layer has a 1×1 filter and one stride. The depth and stride are changed according to the sequence for the parallel block, and extra layers are added before inserting the parallel

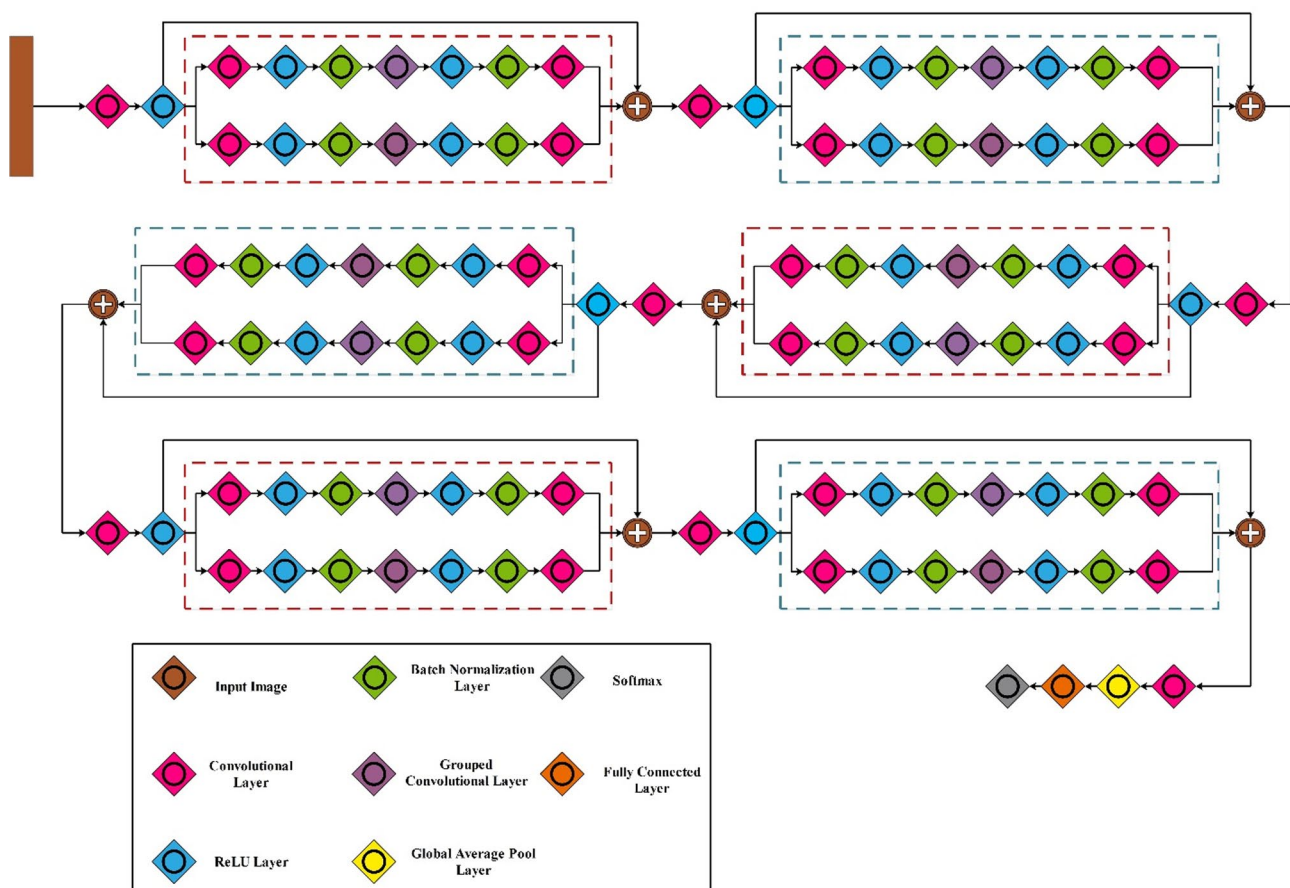


Fig. 6 Proposed layered model structure using 6-block inverted bottleneck design

residual block. A parallel block with an inverted residual bottleneck has two pathways and one skip link.

Next, a convolutional layer was added with a 3×3 filter size, 64 depth, and stride 2. The third residual block follows a consistent layer framework with a 128 and 64-depth size of a convolutional layer. This convolutional layer has a 1×1 filter size with a stride of 1. The depth and stride are changed according to the sequence for the parallel block, and extra layers are added before inserting the parallel residual block. A parallel block with an inverted residual bottleneck has two pathways and one skip link.

Another convolutional layer with a 3×3 filter size, depth of 128, and stride of 2 is added, followed by a ReLU activation layer. The fourth residual block uses an identical layer pattern: a 256-depth convolutional layer, followed by a 128-depth layer with a 1×1 filter size and a stride of (1) Repeat this with the parallel block, modifying the depth and stride, where extra layers are added before the parallel residual block insertion. An inverted residual bottleneck parallel block has two paths and one skip connection. After that, another convolutional layer has been added with a 3×3 filter size, 256 depth, and a stride value of (2) The fifth residual block uses the same layer pattern: a 512-depth convolutional layer, followed by a 256-depth layer with a 1×1 filter size and a stride of 1. Repeat this with the parallel block, modifying the depth and stride, where extra layers are added before the parallel residual block insertion. This block contains two paths and one skip connection.

The following parallel path in the sixth residual block has four layers like the previous block: convolutional, ReLU activation, BN, and grouped convolutional. The sixth residual block contains a convolutional layer of depth size 1024 followed by another convolutional layer of depth size 512-depth, having a 1×1 filter size and a stride of 1. Repeat this with the parallel block, modifying the depth and stride, where extra layers are added before the parallel residual block insertion.

The final convolutional layer uses a 3×3 filter with a 1024 depth and stride of 2. Before neural network classification, a global average pooling layer condenses feature mappings into one channel value. A fully connected layer normalizes the output, followed by a softmax layer for classification. The Categorical cross entropy is the loss function of the proposed model which is defined as $-\sum_{z=1}^N K_z \log(\hat{K}_z)$, where K_z is the labels, $\hat{K}_z$ predicted probioloty of class z , and N is the number of target classes.

Proposed network-level fusion

In the network level fusion, both designed models such as 5-Block Inverted Residual and 6-Block Inverted

Bottleneck are fused using a depth concatenation layer, as shown in Fig. 7. The depth concatenation layer is a type of layer that merges inputs that have identical height and width values along the channel dimension. By employing this particular layer, the network's ability to integrate both low-level and high-level input enhances the accuracy of predictions through the process of concatenating feature maps.

A concatenation layer is used to combine the feature maps based on their dimensions. It is necessary to assume that the feature map is designated as $a_1, a_2, a_3, \dots, a_n$ with the dimension denoted as $h \times w \times c$. The variable h represents the height, and w is denoted as width and c is denoted as channel. So mathematically it work as:

$$Depth_{con}(a_1, a_2, a_3, \dots, a_n) = [a_1, a_2, a_3, \dots, a_n] \quad (8)$$

The feature map dimension, denoted as $h \times w \times (c_1 + c_2 + \dots + c_n)$ in the findings, represents the average depth of all the input feature maps. The subsequent layer in the neural network operates by inputting the concatenation feature map. In both designed models, convolutional layers are added at the end that passed to the depth concatenation layer for the fusion purposes. After that, a global average pool layer has been added that followed by a fully connected layer, and a softmax layer.

Model training After the fusion of the proposed model, training has been performed using the training images set of the selected datasets. Several hyperparameters have been initialized in the training process, such as learning rate, momentum, Regularization factor, epochs, and optimizer. The stochastic gradient descent (SGD) optimizer has been employed for the training, whereas the Bayesian Optimization [50] has been used to initialize these hyperparameters. The values of these hyperparameters are 0.000256 (LR), 0.733 (momentum), and 100 epochs.

Features extraction The trained model is employed, and a testing images set has been used for the test deep features extraction. The global average pool layer has been selected for feature extraction, the the dimension of extracted features are $N \times 1536$, and later passed to the Shallow Neural Network for classification.

Shallow neural network classifiers

The extracted deep features are passed to shallow neural network classifiers for the features classification. The shallow neural network consists of one or two hidden layers that later utilized for the classification. In this work, we used five shallow neural networks for the classification of testing features such as Narrow Neural Networks (N3), Medium Neural Networks (MN2), Wide

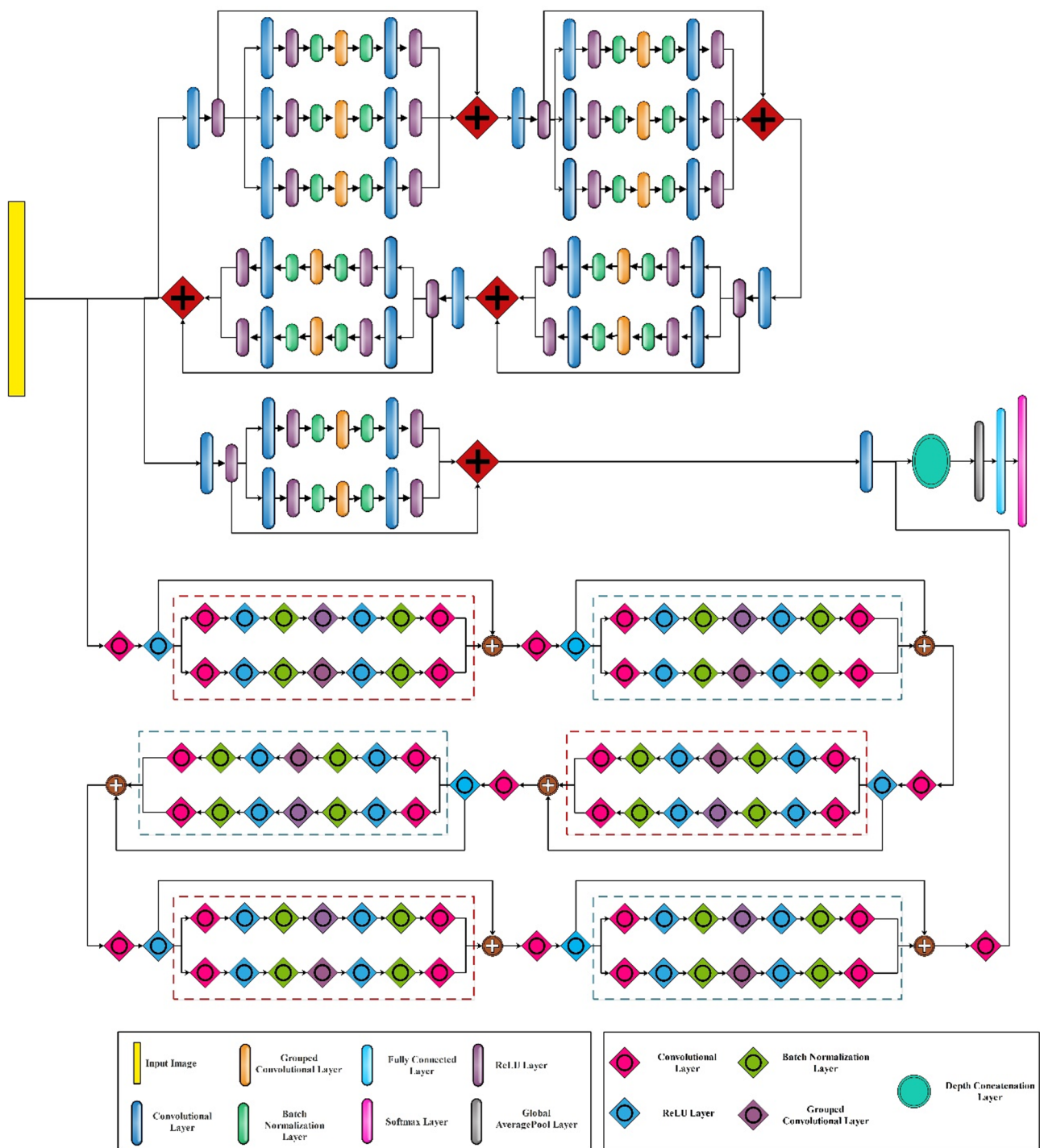


Fig. 7 Proposed network-level fusion architecture of 5-block inverted residual and 6-block inverted bottleneck for skin lesion classification

Neural Networks (WN2), Bi-layered Neural Networks (BN2), and Tri-Neural Neural Networks (TN2). All of these neural networks consists of two hidden layers, one input layer, and one output layer. The hyperparameters of these shallow networks are also optimized using Bayesian Optimization (BO).

Results

Experimental configuration

The proposed system trains with various hyperparameters and evaluates solutions using HAM10000 and ISIC2018, a registered dataset. The datasets are split into 50% training and 50% testing. We start training with 100 epochs, 3000 iterations, 128 mini-batches, and 0.000256

learning rate (LR). Furthermore, a 10-fold cross-validation technique is employed. The performance metrics are accuracy, precision, recall, Area under the curve, f1-score and computation time (sec). these metrics are measured using the neural network classifiers. A recommended configuration for MATLAB 2023b includes a Core i7 CPU with 128GB RAM and a 16GB Graphics Card of RT 3060.

Numerical results

The evaluation of the proposed framework encompass many performance indicators like as accuracy (ACC), time, F1 score (F1-S), area under the curve (AUC), recall rate (ReR), precision rate (PrR), and false-negative rate (FNR). These performance measures are computed for all selected shallow neural network classifiers such as Narrow Neural Networks (N^3), Medium Neural Networks (MN^2), Wide Neural Networks (WN^2), Bi-layered Neural Networks (BN^2), and Tri-Neural Neural Networks (TN^2). Several experiments have been conducted for the validation of the proposed framework such as (i) **Experiment # 1**: Classification results of proposed 5-Block Inverted Residual and 6-Block Inverted Bottleneck models using HAM10000 (enhanced dataset). (ii) **Experiment # 2**: Classification results of proposed 5-Block Inverted Residual and 6-Block Inverted Bottleneck models using HAM10000 augmented dataset (iii) **Experiment # 3**:

Classification results of proposed 5-Block Inverted Residual and 6-Block Inverted Bottleneck models using ISIC2018 (enhanced dataset). (iv) **Experiment # 4**: Classification results of proposed 5-Block Inverted Residual and 6-Block Inverted Bottleneck models using ISIC2018 augmented dataset. (v) **Experiment # 5**: Classification results of proposed network-level fusion on the augmented enhance HAM10000 and ISIC2018 datasets.

HAM10000 classification results

Experiment # 1: In this experiment, the proposed 5-block inverted residual model has been evaluated on enhanced HAM10000 dataset. Table 1(a) presents the results of this experiment. The accuracy of the WN^2 was 82.2% within a time frame of 17.77 s. Additional evaluation metrics encompassed a recall rate of 83.11% and F1-score of 83.01%. The diagonal values for the accurate projected values, as depicted in Fig. 8, are as follows: AKIEC (92%), BCC (83.20%), BKL (66.60%), DF (92.90%), MEL (71.20%), NV (77.00%), and VASC (98.90%). With a time of 17.6080 s, the MN^2 classifier execution time is better.

Table 1(b) presents the classification results of proposed 6-block inverted Bottleneck using the enhanced HAM-10,000 dataset. The WN^2 's greatest accuracy was 77.8% and execution time was 73.246 s. The other calculated measures such as 78.81% recall rate and 78.62%

Table 1 Proposed classification results for HAM10000 dataset using enhanced and augmented data

Classifier	ACC (%)	Time (Sec)	PrR (%)	ReR (%)	AUC	FNR	F1-S (%)
(a) Feature extraction using the Inverted Residual Model (enhanced dataset)							
N^3	77.60	131.20	78.41	78.62	0.90	21.372	78.52085
MN^2	80.70	17.608	81.471	81.671	0.94	18.329	81.57088
WN^2	82.2	17.775	82.92	83.11	0.95	16.89	83.01489
BN^2	77.6	150.80	78.3	78.585	0.91	21.415	78.44224
TN^2	77.0	154.68	77.84	77.842	0.90	22.158	77.841
(b) Feature extraction using the Inverted Bottleneck Model (enhanced dataset)							
N^3	73.20	280.520	73.80	74.21	0.88	25.78	74.0064
MN^2	75.6	47.144	76.128	76.671	0.92	23.329	76.39854
WN^2	77.8	73.246	78.442	78.814	0.93	21.186	78.62756
BN^2	72.8	477.17	73.471	73.8142	0.87	26.1858	73.6422
TN^2	72.4	459.38	73.4	73.3	0.87	26.7	73.34997
(c) Feature extraction using the Inverted Residual Model (augmented dataset)							
N^3	85.30	508.750	85.970	86.04	0.946	13.95	86.0059
MN^2	89.6	44.195	90.10	90.342	0.975	9.658	90.22084
WN^2	88.9	199.32	88.74	89.328	0.979	10.672	89.033
BN^2	85.2	348.81	85.9	86.028	0.94	13.972	85.96395
TN^2	85.1	447.09	85.842	85.985	0.94	14.015	85.91344
(d) Feature extraction using the Inverted Bottleneck Model (augmented dataset)							
N^3	82.70	1023.50	83.510	94.650	0.93	5.350	88.7317
MN^2	86.6	72.104	87.24	87.542	0.96	12.458	87.39074
WN^2	88.1	84.76	88.657	89.085	0.974	10.915	88.87048
BN^2	82.8	832.14	83.671	83.85	0.934	16.15	83.7604
TN^2	82.3	1071.4	83.085	83.3	0.929	16.7	83.19236

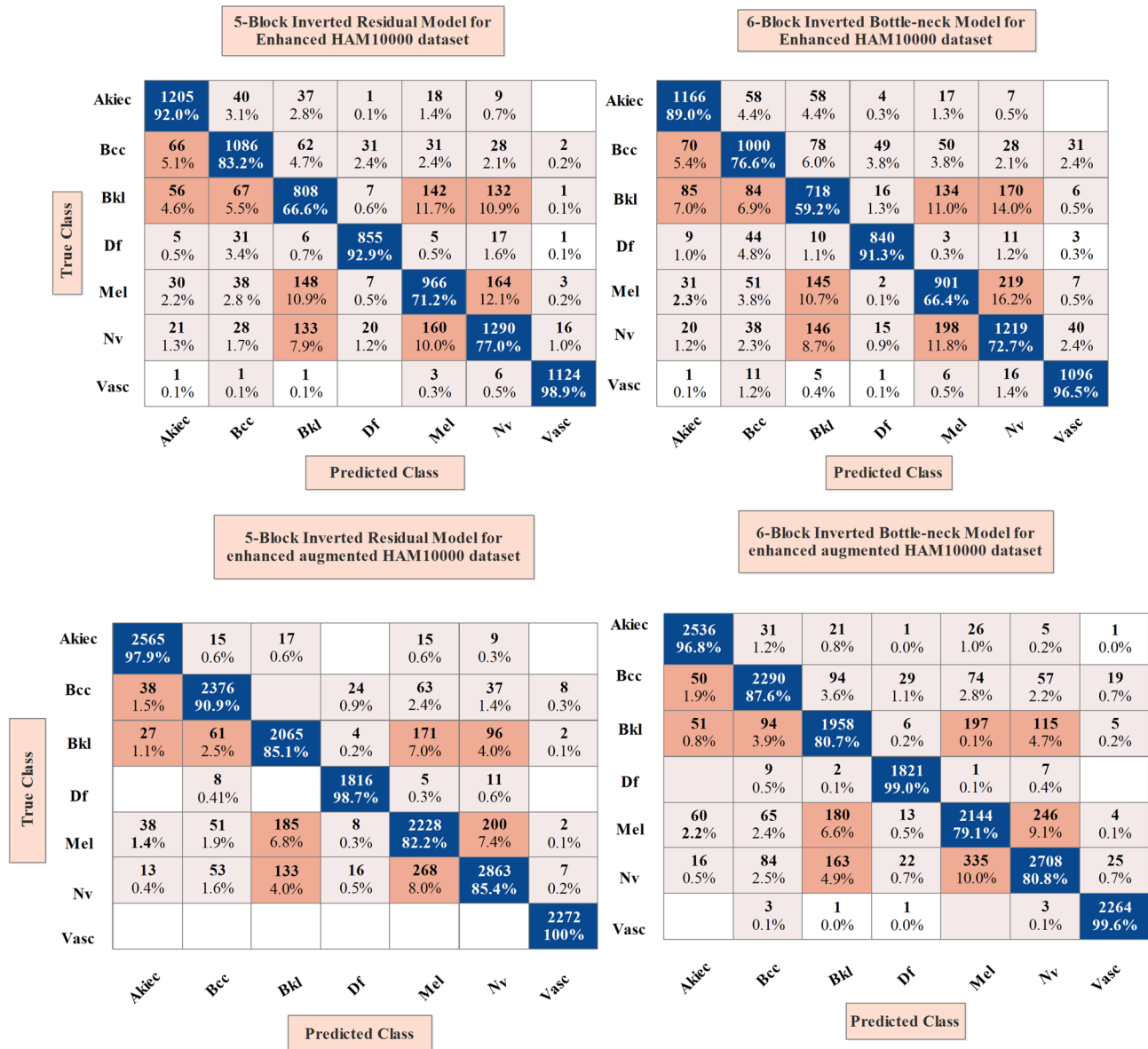


Fig. 8 Confusion matrices and observations of WN^2 using inverted residual model and inverted bottleneck model for HAM10000 dataset

F1-score. Figure 8 presents the confusion matrix of this experiment and it is noted that the prediction accuracy for AKIEC (89%), BCC (76.60%), BKL (59.20%), DF (91.30%), MEL (66.40%), NV (72.70%), and VASC (96.50%). According to the computational time, the MN^2 classifier executes in 47.144 s, which is better than others.

Experiment # 2: Table 1(c) presents the results of proposed 5-block inverted residual model on augmented HAM10000 dataset. For this experiment, the WN^2 achieved the highest performance of 90.7% in a time of 44.723 s. Additional measures include recall rate of 91.45% and F1-score of 91.33%. The confusion matrix of this classifier is also given in Fig. 8 that shows the correct predicted values as follows: AKIEC (97.90%), BCC

(90.90%), BKL (85.10%), DF (98.70%), MEL (82.20%), NV (85.40%), and VASC (100%). Compared the results with Table 1(a), it is noted that the accuracy of the proposed framework is significantly increased for all listed shallow neural networks after the augmentation process.

Table 1(d) presents the classification results of proposed 6-block inverted Bottleneck model and WN^2 classifier obtained the highest accuracy of 88.9% within a time frame of 199.32 s. Additional computed performance measures and their values are recall (89.08%), precision (88.657%), and F1-score (88.87%), respectively. Figure 8 shows the confusion matrix of this classifier that illustrated the correct predicted values such as: AKIEC (96.80%), BCC (87.60%), BKL (80.70%), DF (99%), MEL

(79.10%), NV (80.80%), and VASC (99.60%). Computational time based MN² classifier consumed less time. Compared the results of this table with Table 1 (b), the proposed model accuracy has been increased after the augmentation process. Moreover, the 5-block inverted residual model accuracy is higher (2%) than the 6-block inverted Bottleneck architecture.

Experiment # 3 Similarly to experiment 1 and 2, the experimental process has been conducted on ISIC2018 dataset. Table 2(a) shows the classification results of proposed 5-block inverted residual architecture using enhanced ISIC2018 dataset. The WN² classifier in this table has the highest accuracy of 67.3% in 10.775 s. Figure 9 shows the confusion matrix of this classifier that shows the correct prediction rate of each class such as AKIEC (22.0%), BCC (32.0%), BKL (34.90%), DF (3.90%), MEL (22%), NV (86.20%), and VASC (31.40%). The remaining classifiers accuracy was noted in between 63.7 and 66.2%. Table (b) shows the results of proposed 6-block inverted bottleneck architecture using enhanced ISIC2018 dataset and obtained highest accuracy of 67.3% using WN² classifier. The confusion matrix of this classifier is shown in Fig. 9 that illustrated the correct prediction rate of each class such as AKIEC (22%), BCC (32%), BKL (34.90%), DF (3.90%), MEL (22%), NV (86.20%), and VASC (31.40%).

Overall, the obtained accuracy is not sufficient; therefore, we performed data augmentation on enhanced dataset.

Experiment # 4 Table 2(c) presents the classification results of proposed 5-block inverted residual architecture using ISIC2018 augmented dataset. The WN² achieved the highest accuracy in this table, 91.0% that is improved than the accuracy obtained on the original enhanced dataset (see Table 2(a) results). This classifier has an AUC of 0.97, a recall rate of 90.65%, F1-score of 90.60%, and a FNR of 9.35%, respectively. Figure 9 illustrated the confusion matrix of this experiment that shows the correct prediction rate of each class such as AKIEC (95.80%), BCC (96.40%), BKL (76.70%), DF (100%), MEL (76.70%), NV (89.20%), and VASC (99.880%), respectively. Overall, the prediction rate of each class has been improved after the augmentation process.

The classification of the proposed 6-block inverted bottleneck architecture has been presented in Table 2(d) using augmented enhanced ISIC2018 dataset. The maximum attained accuracy of this experiment is 87.1%, obtained by WN² classifier in 128.12 s. Figure 9 illustrates the confusion matrix of this classifier that show the correct prediction rate of each class such as AKIEC (93.50%), BCC (93.20%), BKL (66.10%), DF (99.90%), MEL (71.60%), NV (83.30%), and VASC (99.30%), respectively. The minimum noted time for this experiment is 106.06 s on MN²

Table 2 Proposed classification results for ISIC-2018 dataset using enhanced and augmented data

Classifier	ACC (%)	Time (Sec)	PrR (%)	ReR (%)	AUC	FNR	F1-S (%)
(a) Feature extraction using the Inverted Residual Model (enhanced dataset)							
N ³	64.10	78.3550	31.21	29.01	0.668	70.98	30.0738
MN ²	66.2	7.2607	35.842	33.114	0.742	66.886	34.42404
WN²	67.3	10.775	36.7	33.84	0.756	66.16	35.21202
BN ²	64.4	96.19	31.128	30.628	0.6777	69.372	30.87598
TN ²	63.5	111.37	28.485	28.5	0.6662	71.5	28.4925
(b) Feature extraction using the Inverted Bottleneck Model (enhanced dataset)							
N ³	58.50	440.210	21.970	22.14	0.626	77.85	22.0556
MN ²	60.1	31.611	23.628	23.585	0.6546	76.415	23.60648
WN²	61.2	34.76	23.1585	23.085	0.7068	76.915	23.12169
BN ²	58.2	351.58	22.657	22.771	0.6315	77.229	22.71386
TN ²	56.9	171.63	19.585	19.814	0.638	80.186	19.69883
(c) Feature extraction using the Inverted residual Model (augmented dataset)							
N ³	88.50	88.0570	88.05	88.15	0.94	11.84	88.1069
MN ²	90.5	89.87	89.87	89.885	0.969	10.115	89.8775
WN²	91.0	90.557	90.557	90.657	0.9797	9.343	90.60697
BN ²	88.7	786.76	88.171	88.285	0.951	11.715	88.22796
TN ²	88.7	801.91	88.214	88.3	0.9559	11.7	88.25698
(d) Feature extraction using the Inverted Bottleneck Model (augmented dataset)							
N ³	85.10	1335.20	84.50	84.720	0.923	15.280	84.6098
MN ²	86.1	106.06	85.5571	85.714	0.952	14.286	85.63548
WN²	87.1	128.12	86.471	86.7	0.965	13.3	86.58535
BN ²	84.8	1221.9	84.2	84.471	0.929	15.529	84.33528
TN ²	84.2	1249.5	83.6	83.828	0.928	16.172	83.71384

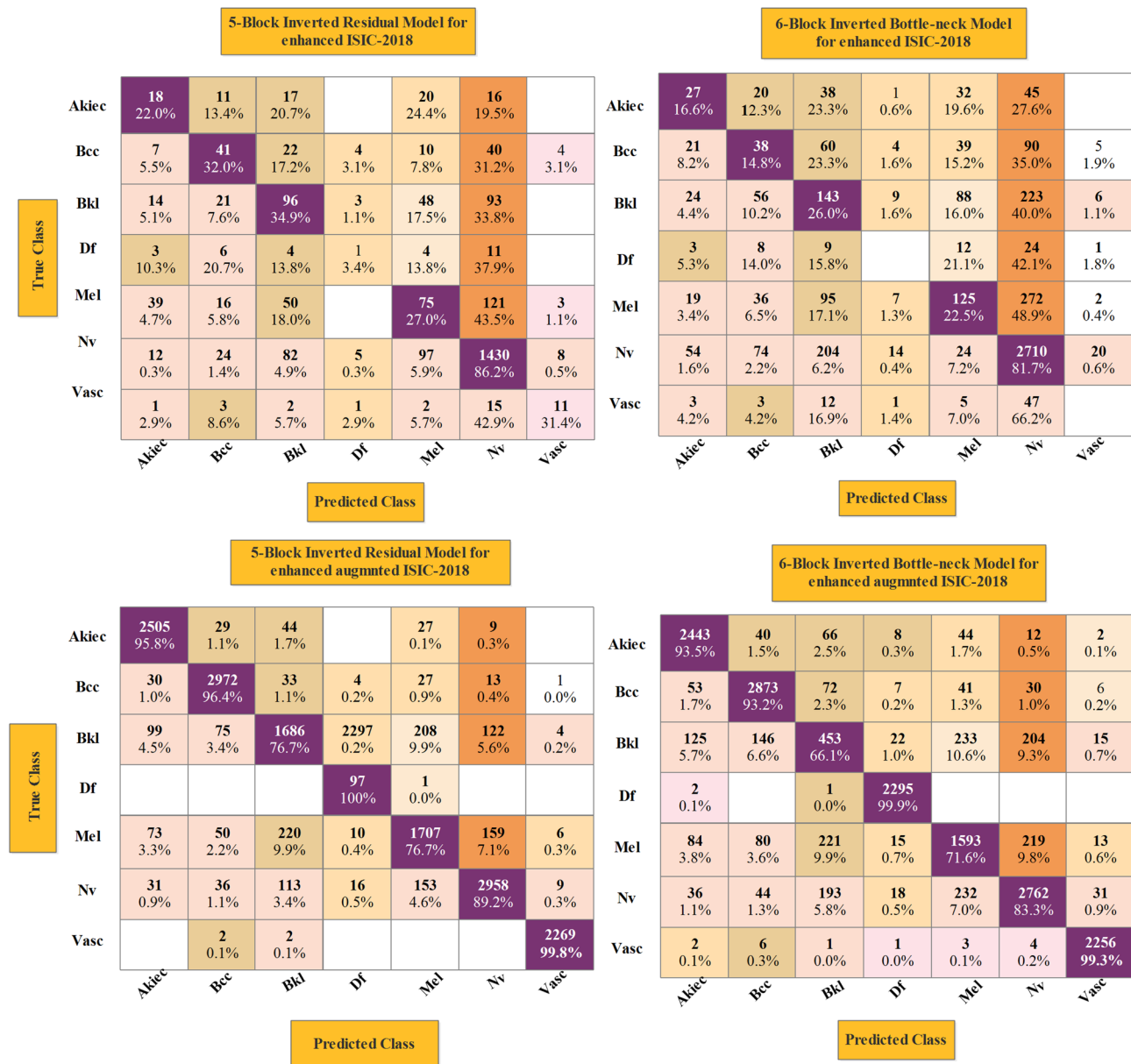


Fig. 9 Confusion matrix and observations of WN^2 using inverted residual model and inverted bottleneck model for ISIC-2018 dataset

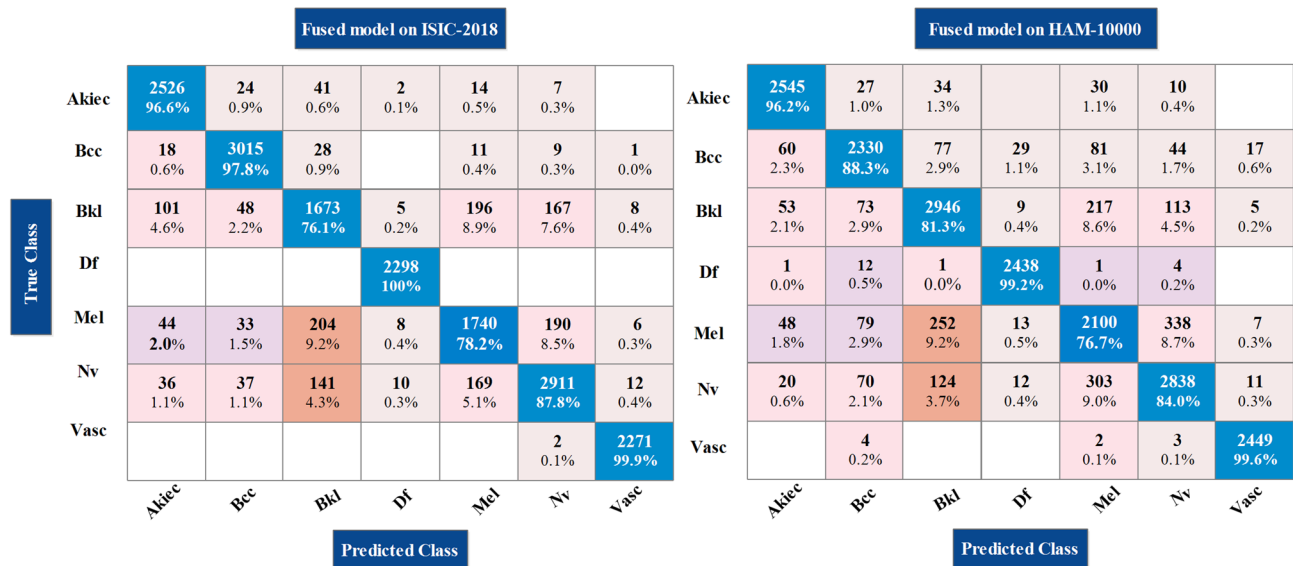
classifier. Compared the results of this table with Table 2 (b), it is show that the proposed architecture improved the accuracy after the augmentation process. Hence, the augmentation process shows the success in improvement in accuracy for the proposed architectures.

Experiment # 5 In this experiment, the proposed network-level fusion has been performed on designed CNN models. The network-level fusion process has been illustrated in Fig. 7. The proposed fusion architecture is tested on enhanced augmented HAM10000 and ISIC2018 datasets. Table 3 (a) shows the results of this experiment for HAM10000 dataset. The WN^2 achieved an optimal performance of 91.3% in 110.27 s. The recall rate of this

classifier is 90.91%, precision rate of 90.91%, and FNR value of 9.086%, respectively. As illustrated in Fig. 10, the predicted value of each class is improved than the confusion matrices presented in Fig. 9. The correct predicted class and values are as AKIEC (96.6%), BCC (97.8%), BKL (76.1%), DF (100%), MEL (72.2%), NV (87.8%), and VASC (99.9%), respectively. Computational time is also noted for each classifier and it is observed that the fusion process not increased the time than individual designed models. Table 3(b) presents the results of proposed fused model on ISIC2018 dataset. The WN^2 classifier achieved an optimal performance of 90.7% in 44.723 s. Recall (91.45%), precision (91.21%), FNR (8.55%), AUC (0.98), and F1-score (91.33%) are obtained. Figure 10 illustrated

Table 3 Proposed classification results after employing the network-level fusion on augmented HAM10000 and ISIC2018 datasets (enhanced augmented)

Classifier	ACC (%)	Time (Sec)	PrR (%)	ReR (%)	AUC	FNR	F1-S (%)
(a) Fusion of both models and extract features on Augmented dataset ISIC-2018							
N ³	90.4	201.9	89.98	90.11	0.963	9.886	90.04495
MN ²	91.2	245.36	90.72	90.87	0.979	9.1286	90.79494
WN²	91.3	110.27	90.91	90.91	0.982	9.086	90.91
BN ²	90.1	294.2	89.71	89.78	0.958	10.215	89.74499
TN ²	89.8	565.7	89.4	89.34	0.960	10.66	89.36999
(b) Fusion of both models and extract features on Augmented dataset HAM-10,000							
N ³	84.0	240.81	89.98	90.11	0.963	9.886	90.04495
MN ²	88.0	153.88	90.72	90.87	0.979	9.1286	90.79494
WN²	90.7	44.723	91.214	91.45	0.983	8.55	91.33185
BN ²	84.7	175.9	89.71	89.78	0.958	10.215	89.74499
TN ²	83.2	162.9	89.4	89.34	0.960	10.66	89.36999

**Fig. 10** Confusion matrix and observations of WN² using proposed network-level fusion on augmented ISIC2018 and HAM10000 dataset

the confusion matrix of this classifier and correct predicted value of each class such as AKIEC (96.2%), BCC (83.3%), BKL (81.3%), DF (99.2%), MEL (76.7%), NV (84.0%), and VASC (99.6%), respectively. Based on all results discussed above, it is noted that the fusion of networks improved the classification accuracy and maintains the computational time.

Analysis and comparison

A detailed analysis of the proposed architecture was conducted based on the precision values for all experiments performed; see Tables 1, 2 and 3. In these tables, the original HAM10000 dataset has achieved a 78.11% precision rate for the inverted bottleneck model. Comparably, this model achieved precision rates of 82.9%, 77.8%, 88.65%, and 88.74% for the enhanced and augmented HAM10000 dataset. The final experiment was the fusion implementation for both models, achieving

the most incredible precision rate of 91.2% on the improved augmented HAM10000 dataset. Similarly, the ISIC2018 dataset precision rate has been analyzed, and a 41.67% maximum precision rate is achieved on the original ISIC2018 dataset. In contrast, 30.62% was the highest precision rate for the inverted bottleneck model. After the augmentation process, the ISIC2018 precision rates improved by 90.91%. A complete visual analysis can be shown in Appendix Section.

After the precision-based analysis of the proposed framework, a comparison has been conducted with several state-of-the-art techniques. Table 4 presents the comparison with SOTA on selected datasets. In this Table [50], used HAM10000 dataset and obtained accuracy of 87.23%. In the next paper [51] used HAM10000 dataset and obtained an accuracy of 86%. In [52], authors used ISIC2018 dataset and obtained accuracy of 90.49%. In [7], authors used ISIC2018 dataset and obtained accuracy of

Table 4 Comparison of the proposed architecture with SOTA

Reference	Year	Dataset	Accuracy (%)
[50]	2022	HAM10000	87.23
[51]	2022	HAM10000	86.0
[52]	2023	ISIC2018	90.49
[7]	2023	ISIC2018	82.5
[53]	2023	ISIC2018	90.11
Proposed	2024	ISIC2018	91.3
		HAM10000	90.7

82.5%. In [53], ISIC2018 dataset was used by authors and obtained accuracy of 90.11%. The proposed architecture

achieved improved classification accuracy of 91.3% and 90.7% for ISIC2018 and HAM10000, respectively.

Lastly, the proposed fused architecture has been passed to explainable AI techniques such as GradCAM and LIME for the model explanation. The fused model is used as a trained architecture, and input testing images have been utilized for the prediction. Figure 11 shows the visual results of this step. The predicted label is compared with the actual label and localized to the important region. The Grad-CAM visualizations in the fourth column indicated what parts of each input image were most influential in the classification decision made by

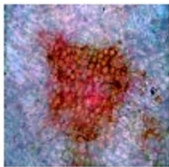
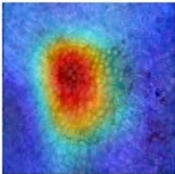
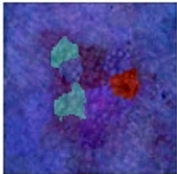
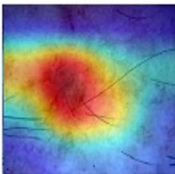
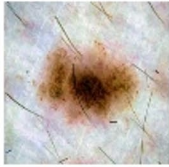
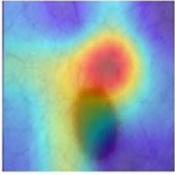
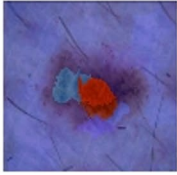
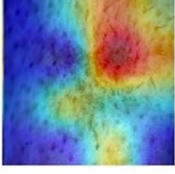
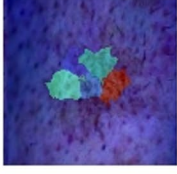
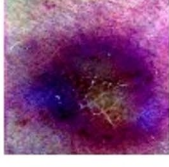
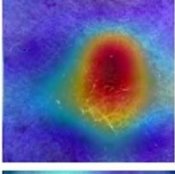
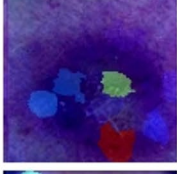
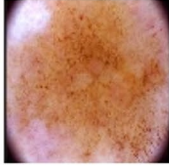
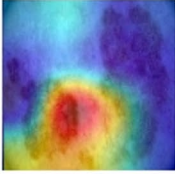
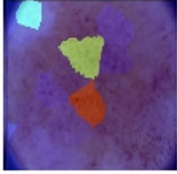
Input Image	Actual Label	Predicted Label	GradCam	Lime
	akiec	✓ akiec		
	akiec	✓ akiec		
	nv	✗ mel		
	bkl	✓ bkl		
	df	✓ df		
	mel	✗ df		

Fig. 11 Explainable AI along with proposed fused architecture based lesion prediction and localization

the model. The heatmap overlayed on the lesion regions utilized a color scale to mark the level of activation from red (the network is highly activating) to blue (the network is minimally activating, or not activating). For the correctly classified cases (akiec, bkl, df), the red regions were all spatially aligned towards the center of the lesion, suggesting the model directed attention to the discriminative features accurately. For misclassifications (nv predicted as mel, and mel predicted as df), the heatmap is more diffuse and in the last column LIME visualizations also decompose the input image into understandable superpixels, and the output indicates which superpixels has the greatest influence based on their directionality towards the final prediction. Each colored area is a group of pixels (superpixels) that supported the predicted label. The each color region (typically red, yellow, blue, cyan or green) are assigned respective positive, neutral contributions such as blue or green. When we look at the correctly classified categorizations, the colored superpixels are closely located around the lesion borders and highlighted the diagnostic features of the lesions. In contrast, all the highlighted segments in misclassified cases potentially even strayed outside the lesion indicating that the model may purely directing its decisions based on sensitive visual cues that were not meaningful to classify the input image.

Discussion

In this section ablation studies are conducted among the pretrained models and proposed models based on number of parameters, layers, size in MB, and input size, as shown in Table 5. According to this table, the proposed models have the lightest architecture from the standard models; with a model size of 14.8 MB and 3.4 M parameters, it is the most compact architecture with relatively few parameters which makes it the most desirable in constrained situations. In contrast, VGG16 is the largest architecture with 138 layers, a huge model size of 528.7 MB, and 138 M parameters. EfficientNetB0

provides a strong compromise between performance and efficiency and has adaptability to the entire range of architectures presented. It has a model size of 20.1 MB and 5.3 M parameters. InceptionV3 and ResNet50 are deeper networks, with model sizes of 92.6 MB and 102.1 MB respectively, and parameters of 23.9 M and 25.6 M respectively. Since these two models are deeper architectures, they require significantly more computing power than EfficientNetB0.

Both proposed IBR⁵Net and IBR⁶Net models have a depth of 101 and 107 respectively, with model sizes of 32.2 MB and 35.5 MB respectively, and 8.4 M parameters and 9.1 M parameters respectively. Thus, it can be suggested that both models feature a more balanced design with a controlled complexity. The proposed network level fused has a significantly higher level of complexity with 210 layers, a model size of 67.3, and 17.5 M parameters. However, even with its greater depth, the fused network remains more compact and efficient in both model size and parameters as compared to earlier models like VGG16 and InceptionV3.

In the next phase, an ablation study has been conducted among the proposed network-level and feature-level fusion process and compared based on the memory utilization, computation time and accuracy, as shown in Fig. 12. Based on this figure, the magnitude of computation time and memory usage are both not preferred; values closer towards the centre are more desirable as they represent less demand on resources. The graph shows that the feature-level fusion occupies more area on the computation time and memory usage aspects showing that feature-level fusion method tends to require more resources. While the network-level fusion process stands below perfect accuracy such as less on computation time and memory usage which implies that network-level fusion might be preferred for efficiency to limited resources. Network-level fusion method attained slightly lower accuracy score, whereas lower performance is significantly offset by minimizing resource consumption.

Conclusion

This paper proposes a deep networks-level fusion architecture along with XAI techniques for skin lesion classification and localization. Contrast enhancement and data augmentation have been performed at the initial stage to improve the training performance of the proposed architecture. Two deep architectures have been designed based on inverted residual and bottleneck fashion that are later concatenated using the network-level fusion technique. Fused architecture is trained on enhanced augmented datasets, while BO is employed to initialize hyperparameters. Results are computed on both datasets and obtained with a maximum accuracy of 91.3% and 90.7%, respectively. A comparison was conducted with

Table 5 Comparison among the proposed and SOTA pretrained model based on basic parameters

Models	Input size	Layers	Model size	No. of parameters
DenseNet121	224×224×3	121	33.2	8.0
MobileNetV2	224×224×3	88	14.8	3.4
InceptionV3	299×299×3	159	92.6	23.9
VGG16	224×224×3	138	528.7	138
ResNet50	224×224×3	177	102.1	25.6
EfficientNetB0	224×224×3	237	20.1	5.3
Proposed IBR ⁵ Net	227×227×3	101	32.2	8.4
Proposed IBR ⁶ Net	227×227×3	107	35.5	9.1
Proposed Fused network	227×227×3	210	67.3	17.5

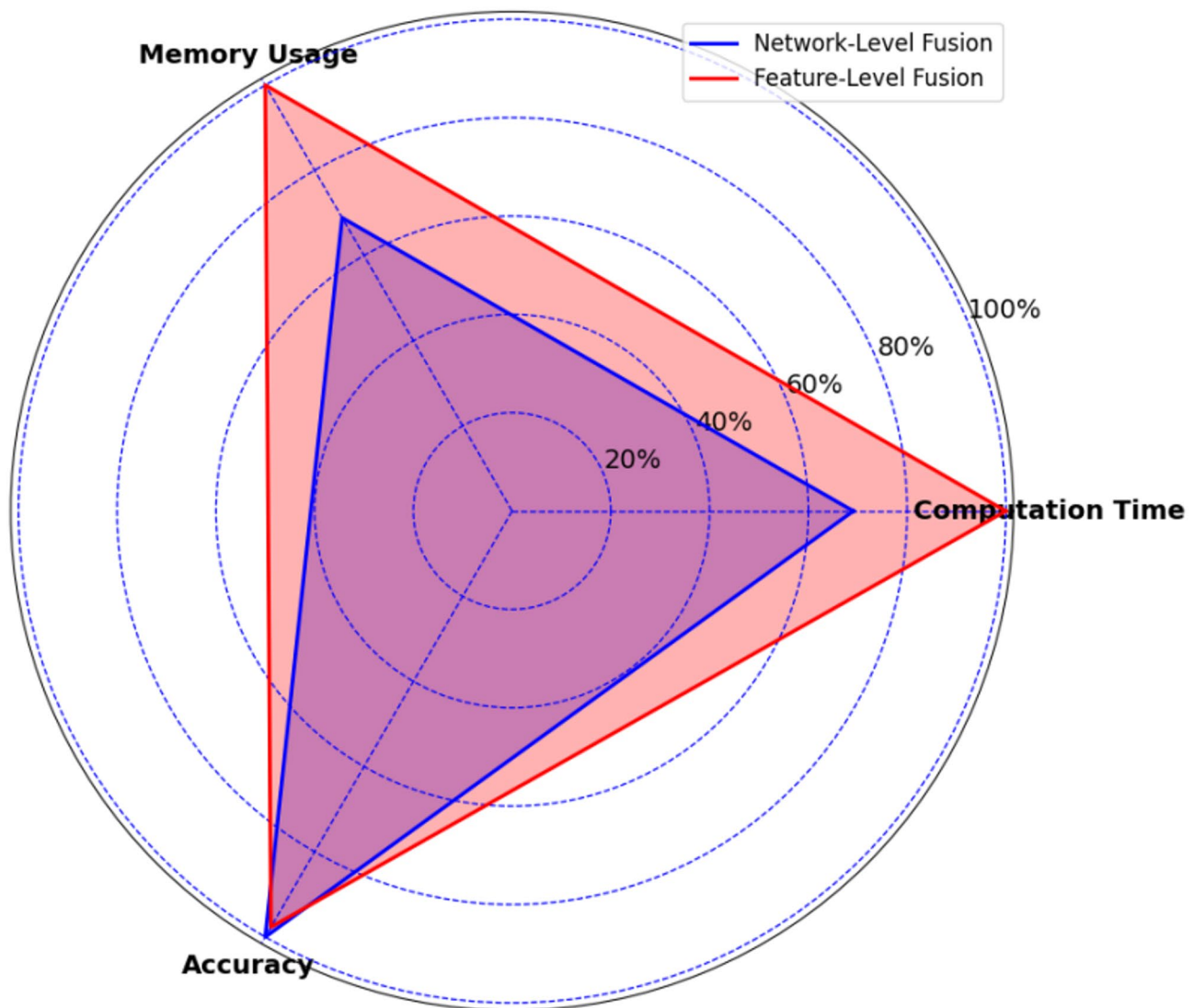


Fig. 12 Ablation study among the feature level fusion and network level fusion based on computation and memory usage parameters

recent techniques, and the proposed architecture performed better. Based on the results, we conclude the following points:

- Contrast enhancement and augmentation process improved the lesion pixel intensity value and learning capability of the designed architectures.
- Fusing bottleneck and inverted residual architectures improved the correct prediction performance and was helpful in model interpretation.
- Network-level fusion architecture controls the total parameters, maintains less computational time, and improves the accuracy and precision rates.

The critical limitation of this network is that it requires more images for the training, which consumes more time and space. In future studies, We will be expanded to

include discussing not only dataset scalability and single-shot learning, but also the possible opportunity of transformer-based architectures, the issuing of light-weight models to running mobile CAD systems, federated learning as a method of privacy-preserving training, and measuring strength against adversarial models.

Supplementary Information

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Supplementary Material 1

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Data availability

The datasets of this work are publicly available for the research purposes. ISIC2018 (<https://challenge.isic-archive.com/data/#2018> and Ham10000 (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>).

Declarations

Ethics, consent to participate, and consent to publish

Not applicable.

Consent for publication

Not applicable.

Relevant guidelines and regulations

Not applicable.

Competing interests

The authors declare no competing interests.

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